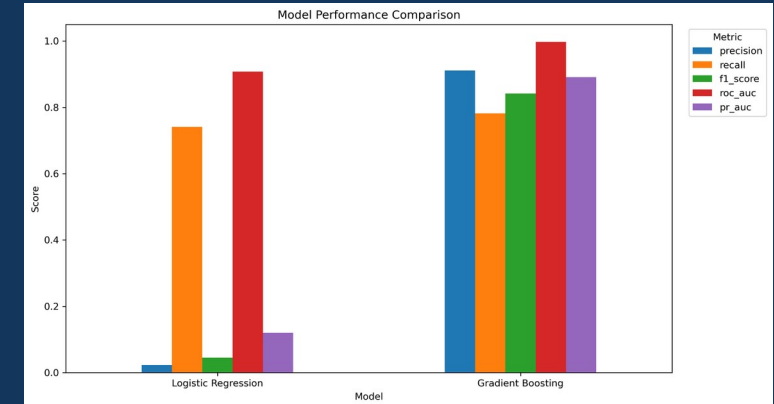


FraudShield BI

Quantifying the Impact of False Positives in Credit Card Fraud Detection

MIS581 Capstone | Business Intelligence and Data Analytics

Atta Febri-Yeboah
Colorado State University Global
Dr. Jamia Mills | July 2026



91.14%

Gradient boosting
precision at 0.50

163

False positives at 0.50

False positives create a hidden “tax” through review workload, customer friction, and missed business opportunities.

Business Problem: Fraud Controls Can Create Customer Friction

The practical challenge

Fraud systems must identify suspicious activity quickly.

Overly aggressive controls can block or review legitimate customers.

False positives can increase manual reviews, call volume, customer frustration, and lost transaction value.

Accuracy alone can hide these tradeoffs in highly imbalanced fraud data.



FraudShield BI reframes fraud detection as a business decision:

Model choice + threshold choice → fraud prevention, customer friction, review workload, and cost.

False Positive Tax

A business-readable estimate of the cost created when legitimate transactions are incorrectly flagged.

Research Questions and Hypotheses

RQ1

How does changing the fraud threshold affect false positives, fraud detection, and estimated False Positive Tax?

H01 rejected**RQ2**

Which transaction characteristics and model outputs are associated with false-positive decisions?

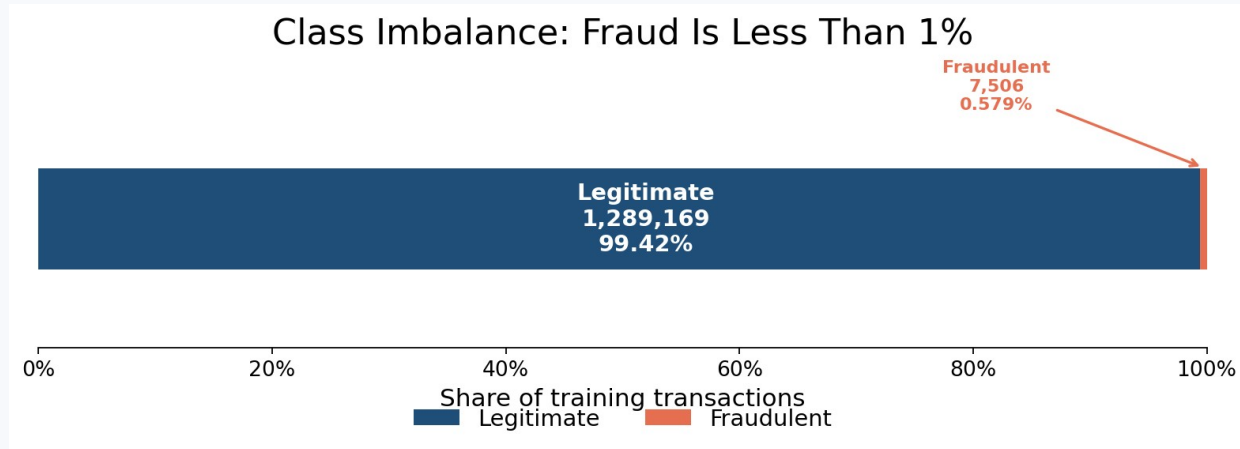
H02 rejected**RQ3**

How can BI dashboards and scenario analysis support fraud-prevention decisions?

H03 rejected

Analytical approach: confusion matrices, threshold analysis, precision-recall metrics, chi-square tests, Mann-Whitney U tests, cost-scenario analysis, and Power BI reporting.

Dataset, Research Design, and Ethical Boundaries

**1,296,675**

training records

555,719

testing records

0.579%

training fraud rate

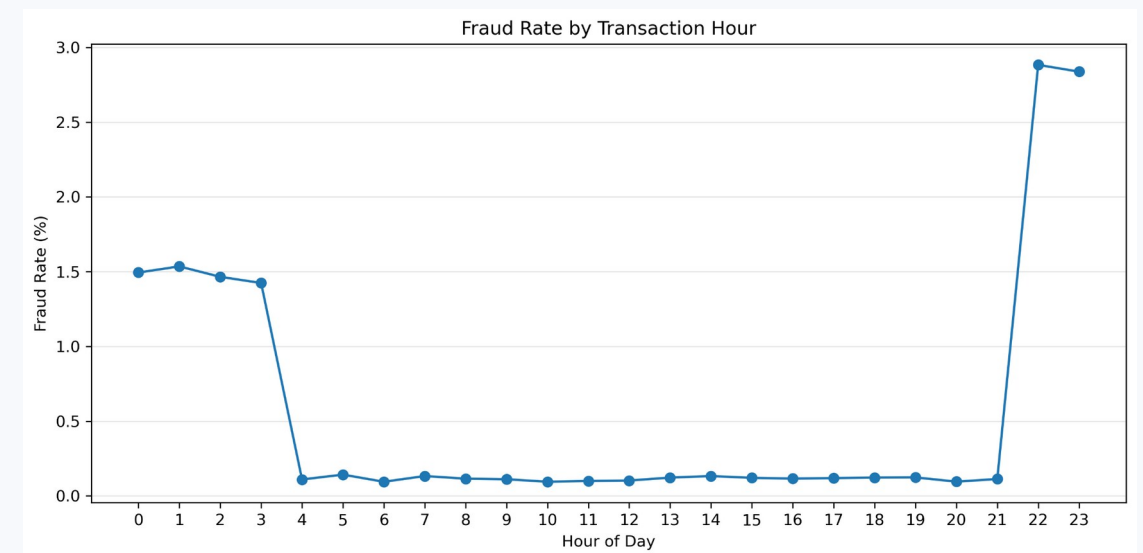
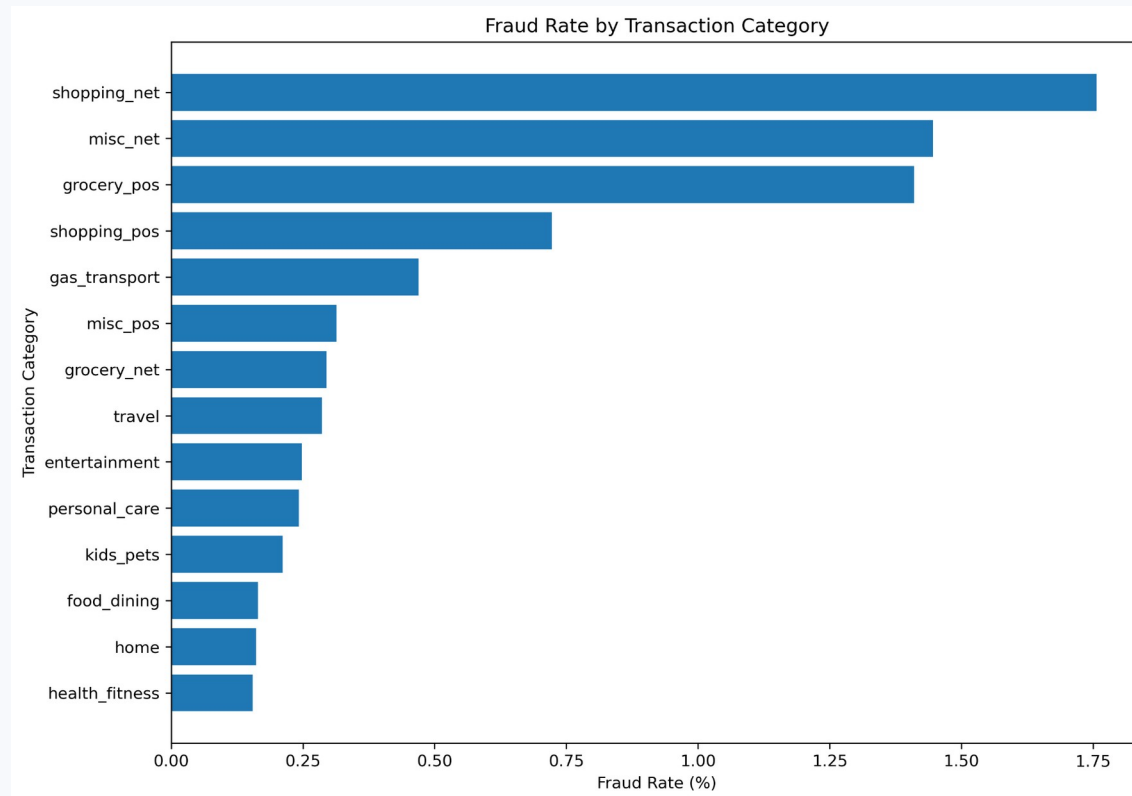
Design choices

Quantitative secondary analysis using a simulated Kaggle/Sparkov transaction dataset.
Pre-split train/test files used; production work should add validation and holdout samples.
Predictors included amount, hour, age, city population, distance, category, and gender.

Ethical guardrails

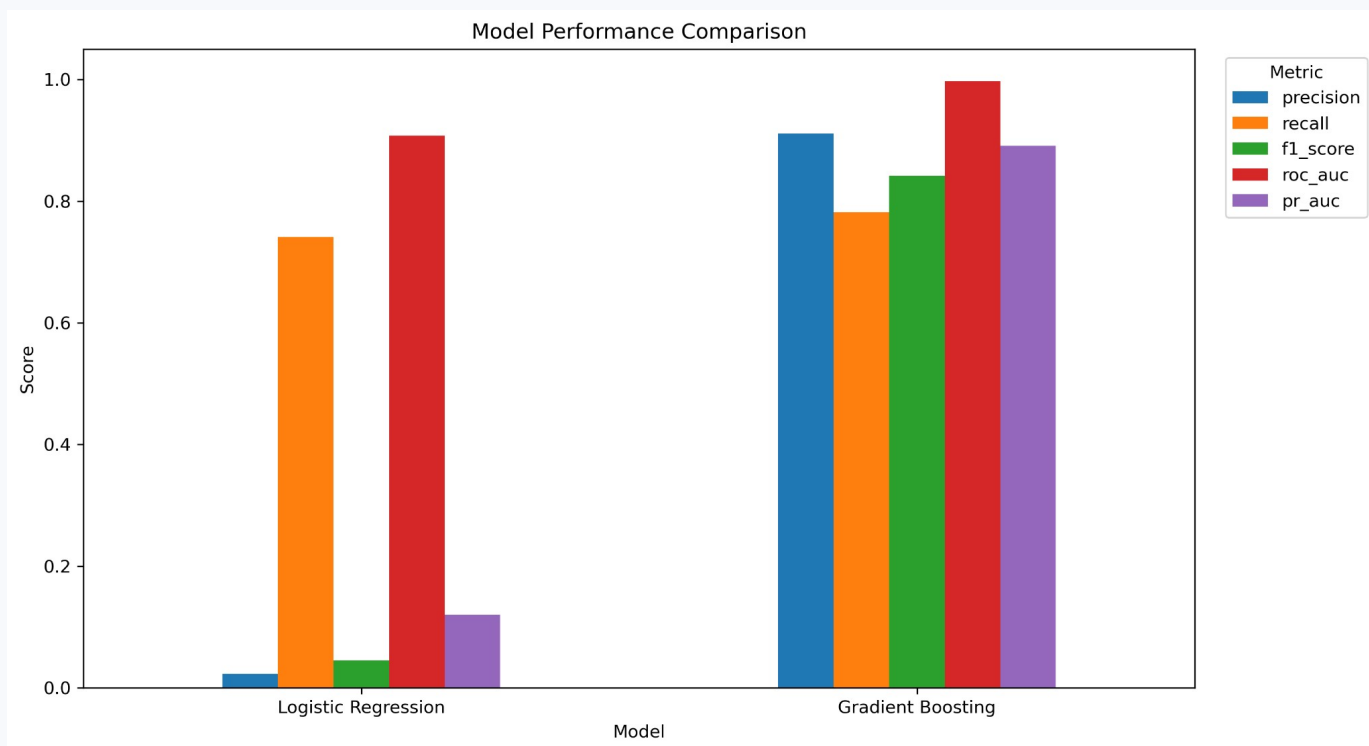
No employer records, customer accounts, or internal fraud rules were used.
Direct identifiers and high-cardinality fields were excluded from modeling.
Dashboard is positioned as decision support, not fully automated production approval.

Exploratory Findings: Fraud Patterns Were Segment-Specific



Key pattern: online shopping, miscellaneous online, and grocery POS had the highest fraud rates. Fraud also rose sharply during late-night hours.

Model Comparison: Gradient Boosting Was the Clear Winner



Default 0.50 threshold

67,204

logistic false positives

163

gradient boosting false positives

2.31%

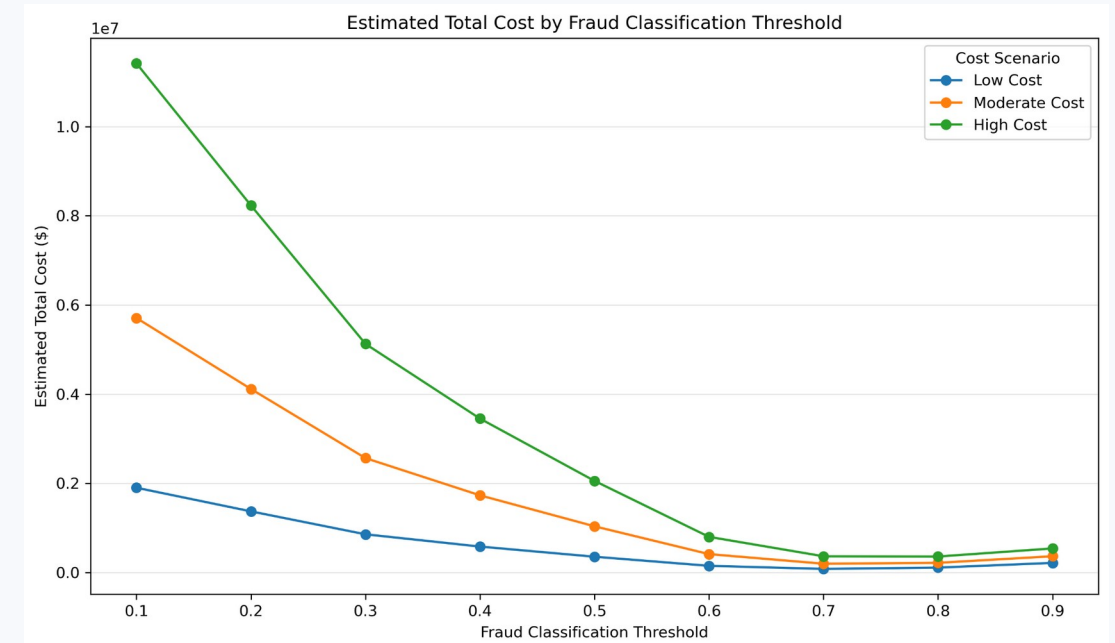
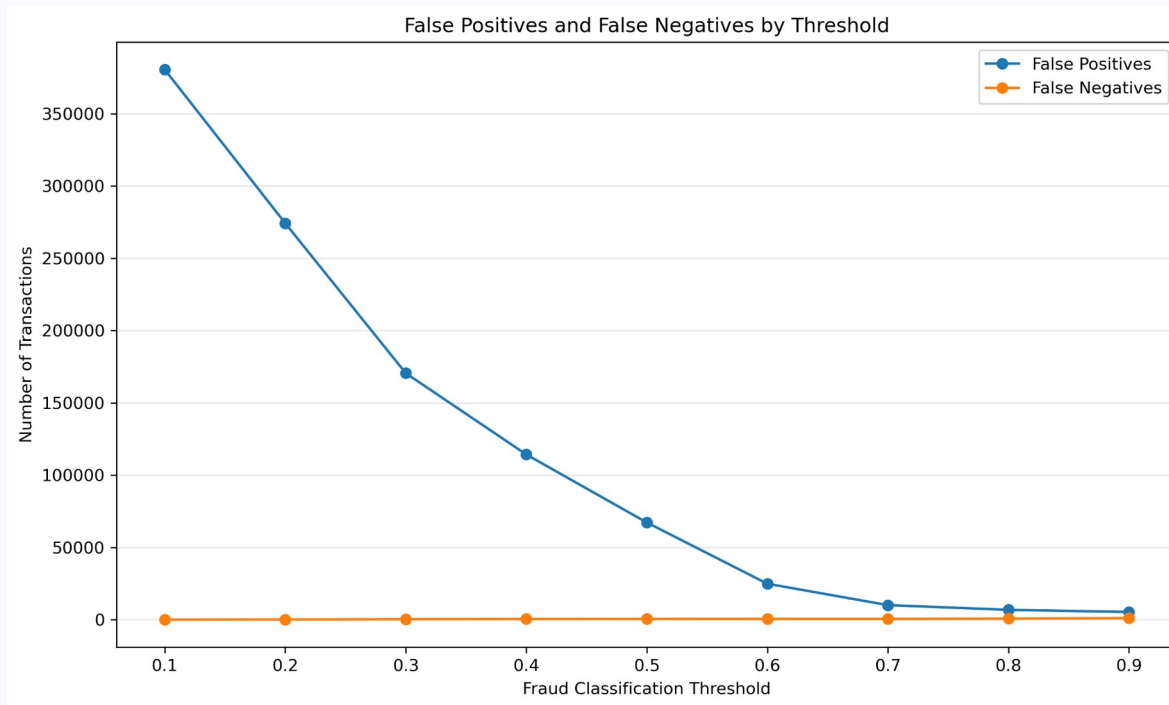
logistic precision

91.14%

gradient boosting precision

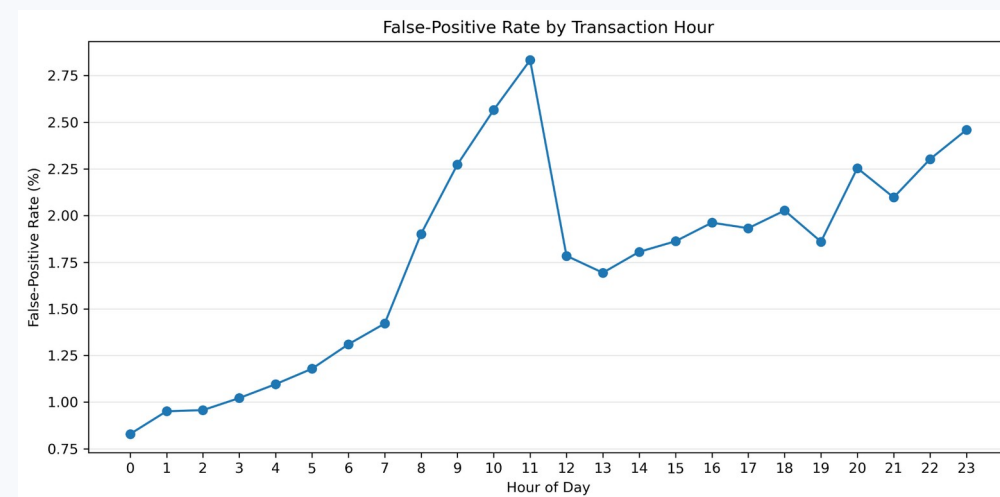
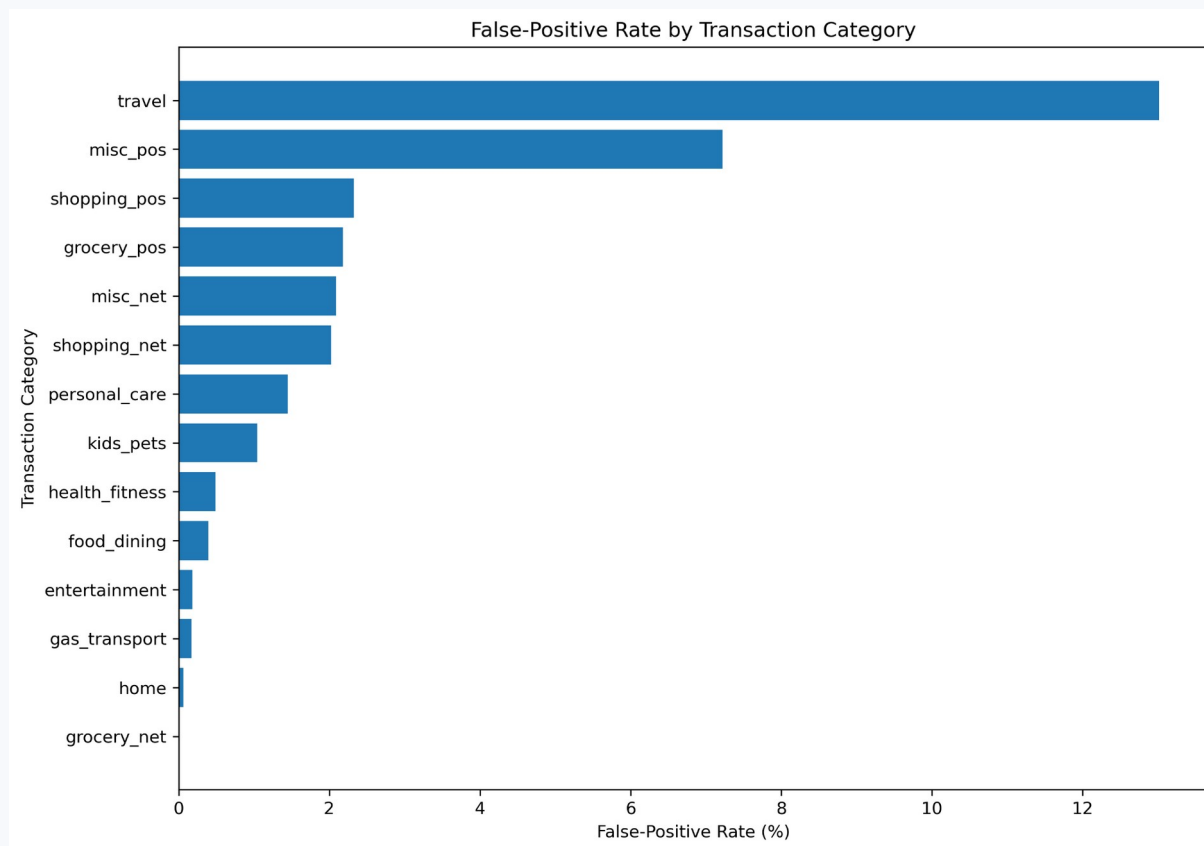
At the same threshold, gradient boosting detected more fraud and reduced false positives by 67,041 compared with logistic regression.

Threshold Analysis: The Cutoff Is a Business Decision



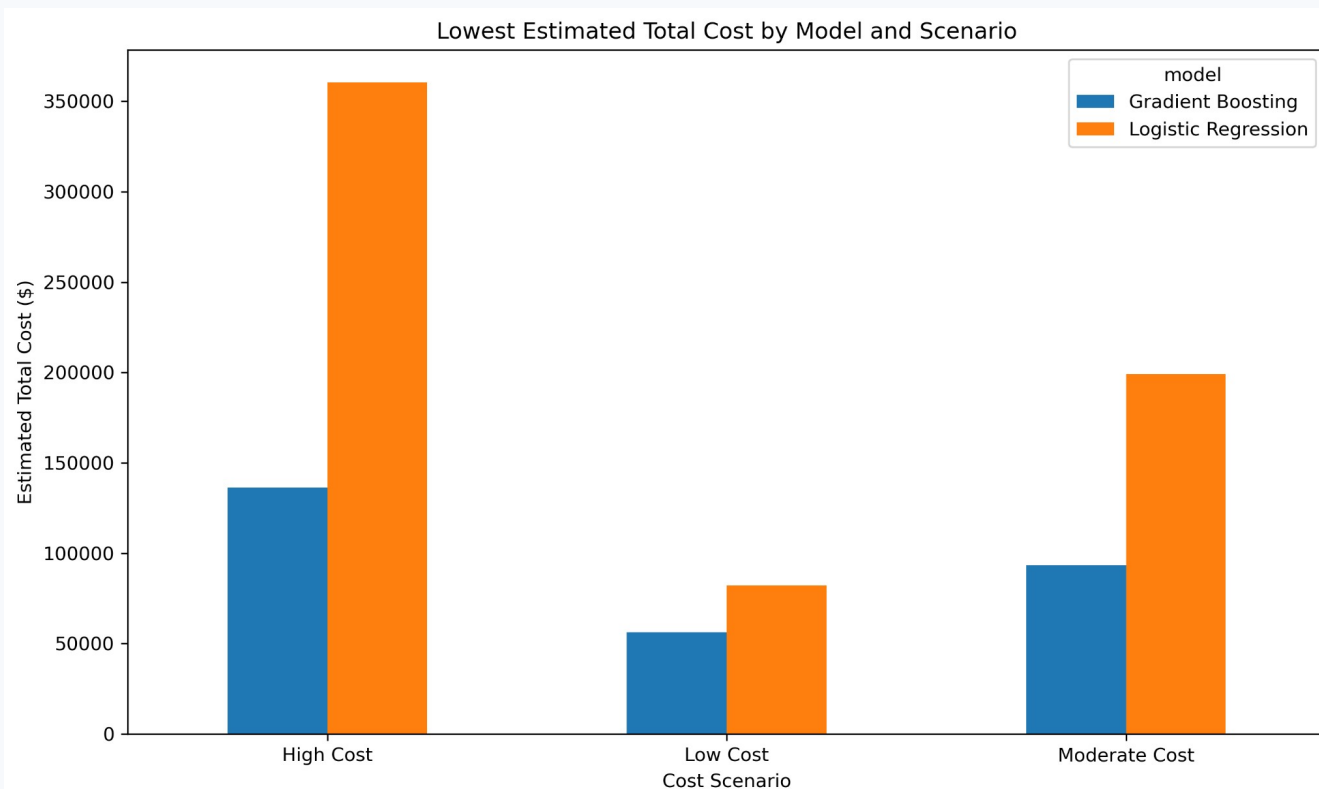
Lower thresholds catch more fraud but can increase false positives. Higher thresholds reduce alerts but may miss more fraud. The preferred threshold depends on cost, customer friction, and review capacity.

False-Positive Impact: The Burden Was Not Evenly Distributed



Operational takeaway: travel had the highest false-positive rate (13.02%), and 11 a.m. had the highest hourly false-positive rate (2.83%). High-impact segments may need step-up authentication instead of automatic decline.

Cost Scenario Analysis: Recommended Model and Threshold



Moderate-cost recommendation

Gradient Boosting
recommended model

0.10

recommended threshold

\$93,392

estimated total cost

1,173

false positives

This recommendation minimizes estimated total cost under the moderate scenario while retaining higher fraud capture than the 0.50 cutoff.

Selected Sources

Bolton, R. J., & Hand, D. J. (2002). Statistical fraud detection: A review. *Statistical Science*, 17(3), 235–255.

Creswell, J. W., & Creswell, J. D. (2023). *Research design: Qualitative, quantitative, and mixed methods approaches* (6th ed.). SAGE.

Custers, B. (2013). Data dilemmas in the information society: Introduction and overview. In B. Custers et al. (Eds.), *Discrimination and privacy in the information society*. Springer.

Dal Pozzolo, A., Caelen, O., Johnson, R. A., & Bontempi, G. (2015). Calibrating probability with undersampling for unbalanced classification. *IEEE Symposium Series on Computational Intelligence*.

Ghosh, S., & Reilly, D. L. (1994). Credit card fraud detection with a neural-network. *Proceedings of the 27th Hawaii International Conference on System Sciences*.

Phua, C., Lee, V., Smith, K., & Gayler, R. (2010). A comprehensive survey of data mining-based fraud detection research. *arXiv preprint*.

Provost, F., & Fawcett, T. (2013). *Data science for business: What you need to know about data mining and data-analytic thinking*. O'Reilly Media.

Kartik2112. (2020). Credit card transactions fraud detection dataset. Kaggle. <https://www.kaggle.com/datasets/kartik2112/fraud-detection>